Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform credit card approval decisions using machine learning, boosting efficiency and accuracy. It tackles manual-review inefficiencies, promising faster operations, reduced risk exposure, and more satisfied applicants. Key features include a machine learning-based credit risk model and near real-time decision-making.

Project Overview	
Objective	The primary objective is to automate the credit card approval process by implementing machine learning classification techniques, ensuring faster and more consistent approval decisions.
Scope	The project comprehensively evaluates applicant financial and demographic data, trains and compares multiple classification models, and deploys the best-performing model as a real-time prediction service.
Problem Statement	
Description	Manual review of credit card applications is slow and inconsistent, leading to processing delays and rejection decisions that vary between reviewers.
Impact	Automating the decision will reduce review time, lower inconsistent rejections, and improve both operational efficiency and applicant satisfaction.
Proposed Solution	
Approach	Employing four classification algorithms — Logistic Regression, Random Forest, XGBoost, and Decision Tree — to analyze applicant data and predict approval likelihood, selecting the best-performing model for deployment.
Key Features	- Implementation of a machine learning-based credit approval prediction model.

	- Real-time decision-making via a Flask web application for instant approval/rejection results. - Cloud deployment through IBM Watson Machine Learning for scalable, continuous scoring.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i3 or above (CPU only, no GPU required)
Memory	RAM specifications	Minimum 4 GB, recommended 8 GB
Storage	Disk space for data, models, and logs	2 GB free disk space
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, XGBoost, pandas, numpy, matplotlib
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm
Data		
Data	Source, size, format	UCI Credit Approval dataset & Kaggle Credit Card Approval dataset, CSV format